

**Exponent Laws with Variable Bases**Question Bank - Additional 3.T1 **ANSWER KEY**

Name

Date

Period

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*Show your work. Circle your final answer.***For questions 1 through 6, write an equivalent expression as a single monomial. Assume no variable equals 0.**

1.  $y^6 \cdot y^7$

$y^{13}$

2.  $(k^3)^8$

$k^{24}$

3.  $(5x^2)^3$

$125x^6$

4.  $\frac{b^{11}}{b^4}$

$b^7$

5.  $(-3a^5)(-4a^2)$

$12a^7$

6.  $\frac{18x^6y^4}{6x^5y}$

$3xy^3$

**For questions 7 and 8, apply more than one law. Write each answer as a single monomial.**

7. Write an equivalent expression.

a.  $(3a^4b^2)(6ab^3)$

$18a^5b^5$

b.  $(2p^3q^5)^3$

$8p^9q^{15}$

c.  $(xy^3)^2 \cdot x^4y$

$x^6y^7$

8. Write an equivalent expression.

a.  $\frac{(2x^3y)^4}{8x^7y}$

$2x^5y^3$

$16x^{12}y^4$  over  $8x^7y$ .

b.  $\frac{m^5n^6}{m^2n^6}$

$m^3$

$n^0 = 1$ .

c.  $\frac{(5c^2)^2}{5c^3}$

$5c$

$25c^4$  over  $5c^3$ .

**Working backwards.**

9. Find the value of  $n$  that makes each statement true.

a.  $a^n \cdot a^5 = a^{12}$

$n = 7$

b.  $(x^3)^n = x^{18}$

$n = 6$

c.  $\frac{y^n}{y^4} = y^6$

$n = 10$

$n - 4 = 6$ .

10. A rectangle has length  $9a^5$  and width  $3a^2$ .

a. Write its area as a single monomial.

$27a^7$

b. A square has side length  $7b^3$ . Write its area as a single monomial.

$49b^6$

11. Which expression is equivalent to  $(4y^2)^3$  ?

☐ A.  $12y^6$

☐ B.  $4y^6$

☒ C.  $64y^6$

☐ D.  $64y^5$

**C**

12. Select ALL of the expressions that are equivalent to  $m^{10}$  .

☒ **A.**  $m^3 \cdot m^7$

☒ **B.**  $(m^2)^5$

☐ **C.**  $m^2 \cdot m^5$

☐ **D.**  $\frac{m^{20}}{m^2}$

☒ **E.**  $\frac{m^{14}}{m^4}$

☒ **F.**  $(m^5)^2$

***A, B, E and F***